Case Report

Numerical Investigation of the Mechanical Behavior of Polyethylene Beams Containing Recycled Polyethylene

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ABSTRACT

There are many types of materials, one of which is plastic materials, and among the common plastic materials is polyethylene. These materials play an important role in everyday use because they have good properties such as strength, flexibility, lightness, low cost, and better thermal properties and electrical insulation, which can be used in many industries. To enhance the performance of polymers, they are often combined with other materials to form polymer composites consisting of two or more phases, at least one of which is polymeric. By blending a polymer with another polymer or reinforcing material, unique combinations of mechanical and physical properties can be achieved. In recent years, the increasing consumption of plastics has intensified concerns regarding resource management, environmental sustainability, and plastic waste generation, leading to greater interest in recycling and reintroducing these materials into the production cycle. Characteristics such as environmental compatibility, low density, durability, favorable life-cycle performance, and cost-effectiveness have contributed to the widespread use of plastics and polymer composites in various engineering applications. In this study, virgin polyethylene was used as the matrix material and reinforced with recycled polyethylene at different weight ratios, including a specimen composed entirely of recycled polyethylene. The objective of this research was to determine the optimal recycled polyethylene content for practical applications while maintaining suitable mechanical performance. Based on the simulation and mechanical analysis results, the composite containing 30% recycled polyethylene exhibited the best balance between mechanical strength, dimensional stability, and reduced consumption of virgin material.

Introduction

Plastics are a vital class of materials widely used across various industries due to their favorable mechanical properties, including strength, flexibility, light weight, low cost, and ease of processing, as well as their effective thermal and electrical

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insulation and aesthetic versatility [1]. The increasing use of these materials has resulted in numerous studies to understand their mechanical behavior [2,3]. Their widespread application has led to production levels that surpass most man-made products, except for cement and steel in the construction sector [4]. However, the production and incineration of plastic waste emit about 400 million tons of carbon dioxide annually. Consequently, the increase in plastic production led to a significant volume of plastic waste. Unfortunately, their significant negative effects on ecosystems and environmental footprints have not reduced their growing popularity [5]. Plastic waste contains polymers that do not decompose easily due to biological factors such as bacteria, fungi, and yeasts or environmental factors such as light, heat, moisture, oxygen, and chemicals; these materials take many years to break down and often include added chemicals to enhance properties and lower costs [6].

Polymer composites are materials composed of two or more distinct phases, in which at least one phase is a polymer, and the other phases may consist of fibers, particles, or fillers. Combining polymers with materials such as glass, carbon, or other polymers can produce composites with enhanced and tailored properties. Common synthetic polymer composites include glass-, carbon-, and polymer-fiber-reinforced materials, thermoplastics, thermoset resins, and polymer blends. These composites are mainly developed to improve properties such as strength, stiffness, toughness, and dimensional stability through the incorporation of fibers or particles into a polymer matrix. Additionally, the use of low-cost and readily available fillers allows the production of materials suitable for a wide range of applications [7]. Due to their lightweight nature and high specific strength and stiffness, composite materials are widely used in aerospace, automotive, wind turbine, marine, and structural applications. Plastics are also extensively utilized in products such as food packaging, furniture, containers, pipes, automotive components, and beverage bottles. While thermoset plastics are generally non-recyclable, thermoplastics can be recycled. Large amounts of plastics, including polypropylene, polyethylene, and polyethylene terephthalate, are discarded annually, creating major environmental concerns. Conventional disposal methods such as incineration and landfilling can lead to pollution, harmful gas emissions, and groundwater contamination, posing risks to both human health and ecosystems. Recycling is considered one of the most effective approaches for managing plastic waste, involving collection, separation, cleaning, and reprocessing into reusable materials or new products. The primary recycling methods include mechanical and chemical recycling [8]. The urgent need to reduce the environmental impact of plastic waste has attracted increasing attention from both researchers and society [9]. Among recycling methods, mechanical recycling is particularly promising because it is cost-effective, scalable, solvent-free, and applicable to many types of polymers [10,11]. In addition to reducing waste, this process supports sustainability by promoting the reuse of existing plastic materials and minimizing the overall environmental footprint [12,13].

Fantozzi *et al*. [14] investigated the behavior of recycled plastic and extracted its material properties for the design of structural and non-structural elements. In this study, three stiffened sheets were made of 80% recycled plastic (about 50% recycled polypropylene (rPP) and about 50% high-density polyethylene (HDPE) with some low-density polyethylene (LDPE) and 20% virgin polypropylene copolymer (PP). Three-point bending tests were performed on three samples. The material properties were extracted using the Ramberg-Osgood equation and after implementation in finite element models, it was observed that the simulated material shows similar behavior to that recorded during the experiment. The results can be used to study the design of a structural or non-structural component containing recycled plastic. In other research, Pešić *et al*. [15] investigated the benefits of simply extruded recycled HDPE plastic fibers in structural concrete. The tensile strength and flexural modulus (rupture) were slightly increased in the presence of HDPE fibers by 3–14%. The fibers contributed mainly by providing flexural ductility after cracking and by improving the performance properties of the concrete, such as reducing plastic shrinkage

cracking, drying shrinkage, and water permeability. Singh *et al*. [16] have reviewed the developments of recycled thermoplastic matrix composite materials. These studies focused on the effect of fiber type and content. Additionally, several factors affecting the overall mechanical performance of recycled thermoplastic matrix composites have been investigated, including material mixing, filler content, processing methods, fiber surface treatment, and processing conditions. Previous studies have mainly focused on evaluating mechanical properties such as tensile strength, impact resistance, adhesive strength, and flexural behavior. In another study, Nabih *et al*. [17] developed plastic fibers (PF) from recycled plastic waste and produced woven fibers with characteristics similar to carbon fiber reinforced polymer (CFRP). Compared with CFRP manufacturing, the production process and cost of these recycled fibers were significantly lower. Experimental results demonstrated that the fabricated fibers exhibited mechanical behavior comparable to CFRP. In particular, the ultimate load capacity of beams reinforced with PF and CFRP increased by approximately 45.45% and 51%, respectively. This improvement was attributed to the ability of both fiber types to limit crack propagation in the beam bending region. Furthermore, the type of reinforcement influenced the failure mode: beams reinforced with polyethylene fibers exhibited a gradual and ductile failure, whereas beams reinforced with CFRP showed a brittle and sudden failure.

Singleton *et al*. [18] fabricated a composite laminate based on natural flax fibers and recycled heavy-duty polyethylene. The mechanical properties of the composite were demonstrated under tensile and impact loading. A significant increase in the toughness of the composite was qualitatively demonstrated based on the main deformation and fracture mechanisms identified by optical microscopy and scanning electron microscopy. Improvements in strength and stiffness along with high toughness can also be achieved by varying the fiber volume fraction and controlling the bonding between the composite layers. Titone *et al*. [19] examined landfilling and incineration or composting of recycled materials, and explored management options that had not been considered. The study explores new solutions and insights that could revolutionize polymer waste management and advance more sustainable recycling practices. This is important in the current context of plastic pollution, which is a major issue, and the scientific community is focusing its efforts on recycling these materials. Berriet *et al*. [20] investigated the effects of the recycling method on the mechanical behavior of polypropylene (PP)/elastomeric particles, talc, filled or not, through different loading tests such as tensile, flexural, and creep tests. The tensile and flexural tests led to the evolution of some similar properties such as elastic modulus values. Both materials have a relatively finite macroscopic strain behavior, so that by considering each component (talc, EPDM, or EPR elastomeric part), the intrinsic behavior can be well investigated in the mechanical model. In this study, beam specimens were designed to investigate the effect of different proportions of recycled polyethylene blended with virgin polyethylene, including a specimen made entirely of recycled polyethylene. The mechanical behavior of the samples was then evaluated in terms of stress, displacement, and generated plastic strain. The incorporation of recycled polyethylene provides an effective approach for reusing plastic waste, considering its long decomposition time and the predictable mechanical performance of recycled materials. Furthermore, an attempt was made to identify the optimal recycled polyethylene content for various applications to reduce material costs while maintaining the required performance and efficiency. It is noteworthy that the innovation of this research lies primarily in the combination of simulation and the creation of recycled materials. Focusing on simulation and its application in the development of recycled materials will play a significant role in future material supply.

Materials and Methods

Materials

Virgin HDPE has a density of 0.941 to 0.965 and is a thermoplastic material composed of carbon and hydrogen atoms bonded together to form

high molecular weight products [21]. The mechanical properties of the other composites in this research were extracted based on previous papers, which are presented in Table 1.

Table 1: Mechanical properties of the samples studied in the present study [22,23]

Composition ratio (v-r) in %	E (MPa)	$\sigma_1(\varepsilon=0)$	$\sigma_2(\varepsilon=0.05)$	$\sigma_3(\varepsilon=0.1)$	ρ (g/cm³)	Poisson's ratio
100-0	1,100	12.58	18.6	20.67	0.95	0.42
70-30	1,040	11.04	16.34	18.16	0.93	0.42
50-50	1,000	5.67	11.41	12.68	0.92	0.42
30-70	960	5.11	9.17	10.19	0.91	0.42
0-100	900	4.45	6.96	7.73	0.9	0.42

In Table 1, three points of the plastic stress-strain curve are presented so that the elastic and plastic analysis can be performed simultaneously. The modulus of elasticity and yield stress of the composite material can also be estimated using the rule of mixtures.

In the Equations 1 and 2, wr is the weight percentage of recycled polyethylene and wv is the weight percentage of virgin polyethylene, while Er and Ev are the elastic modulus of virgin and recycled polyethylene, respectively. Additionally, sigma yr and sigma yv are the yield stress of recycled and virgin polyethylene, respectively, which lead to the elastic modulus and yield stress of the composite (blend).

$$w_r \times E_r + w_v \times E_v = E_{blend} \quad (1)$$

$$w_r \times \sigma_{yr} + w_v \times \sigma_{yv} = \sigma_{yblend} \quad (2)$$

Methods and Simulation

The Abaqus Software was used for software simulation. In this section, standard beam specimens with dimensions of 150 × 20 × 5 mm, based on the ASTM D790 standard, were considered for elastic and plastic bending analysis under applied loads of 10 and 20 N. These load values were selected to ensure that the specimens entered the plastic deformation range at the higher loading condition. The material properties described in the previous section were implemented in the simulation software, and the applied load was introduced at the midpoint of the beam. The support conditions were defined according to a three-point bending configuration. For accurate numerical analysis, C3D8 elements (in Abaqus, the C3D8 element is an 8-node linear three-dimensional brick element commonly used for solid mechanical and structural analysis) with a mesh size of 1 mm were used. The simulations were performed using a general static analysis step with a duration of 1 s (Figure 1).

Results and Discussion

To verify the work done, a tensile test sample according to the standard, considering the mentioned properties, was subjected to tensile simulation for a beam with new polyethylene. As a verification step, the simulated ultimate stress for virgin HDPE was compared with the value reported by Tesfaw *et al.* [22] (39.98 MPa).

Von Mises Stress

The results (Figures 2-4) showed that with increasing percentage of recycled polyethylene, the maximum Von Mises stress decreased (from 6.88 MPa to 5.20 MPa under 20 N loading). This stress reduction means an increase in the safety factor for the yield stress; however, due to the decrease in the modulus of elasticity, the amount of displacement and deformation of the beam increases. Therefore, the selection of the optimal percentage depends on the design criteria: in applications where mechanical stability and minimal deformation are important, the pure polymer is superior, while for applications where only the prevention of yielding is considered, a higher percentage of recycled material has a more suitable performance.

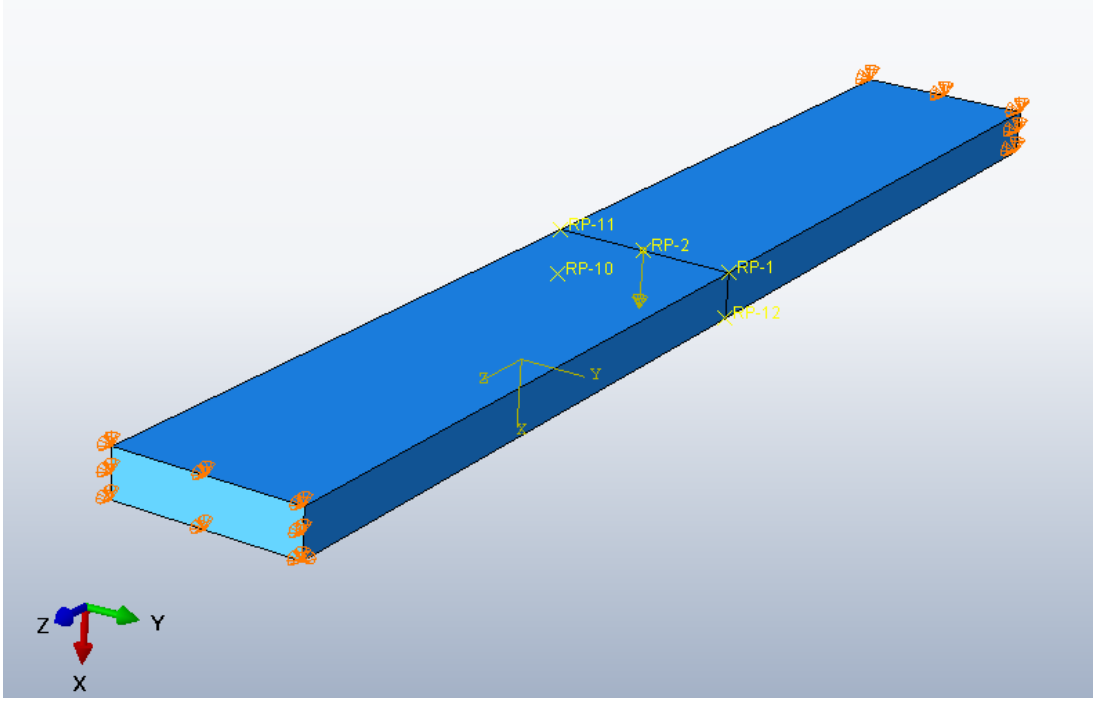


Figure 1: 3D Model designed in the software (Computer-Aided Engineering-CAE).

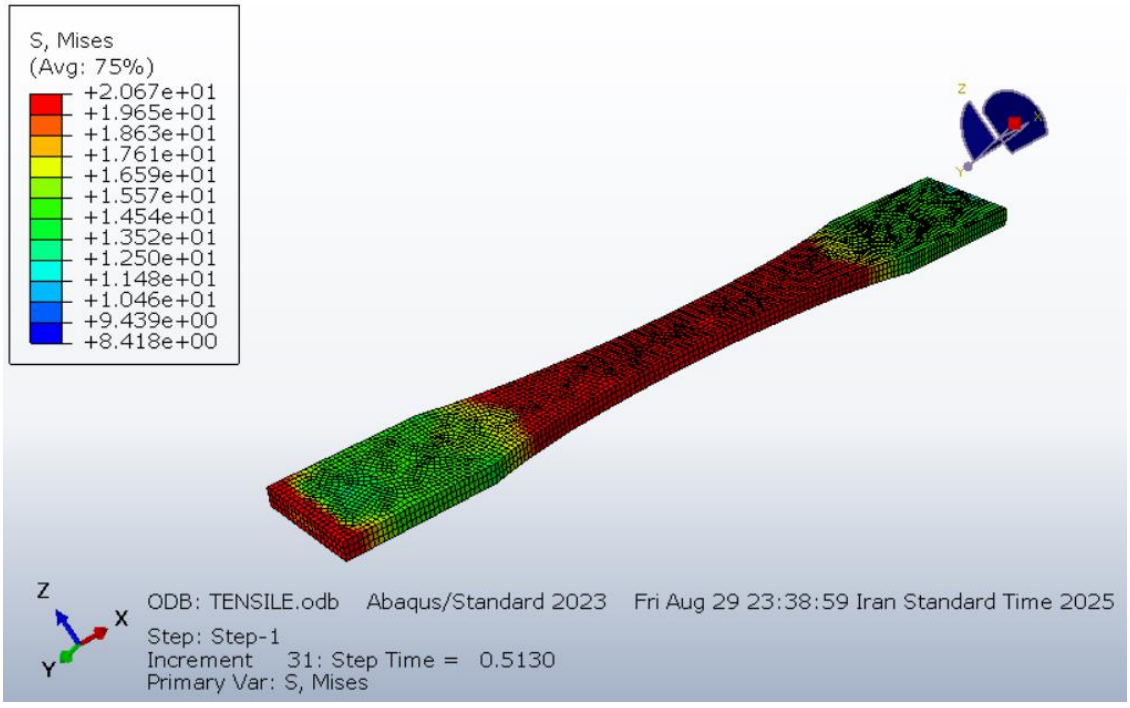


Figure 2: Von Mises stress for tensile testing in CAE software after simulation

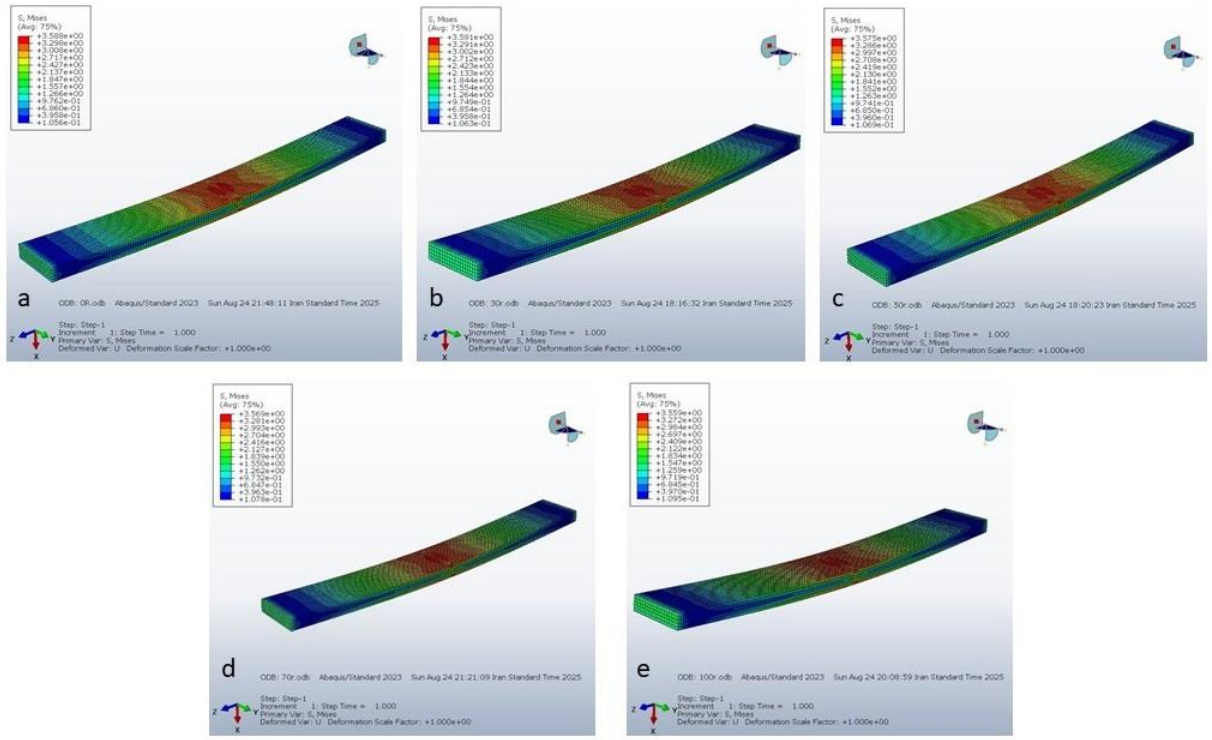


Figure 3: Von Mises stress results under a 10 N applied force for different material compositions: (a) 100% virgin, (b) 30% recycled, (c) 50% recycled, (d) 70% recycled, and (e) 100% recycled

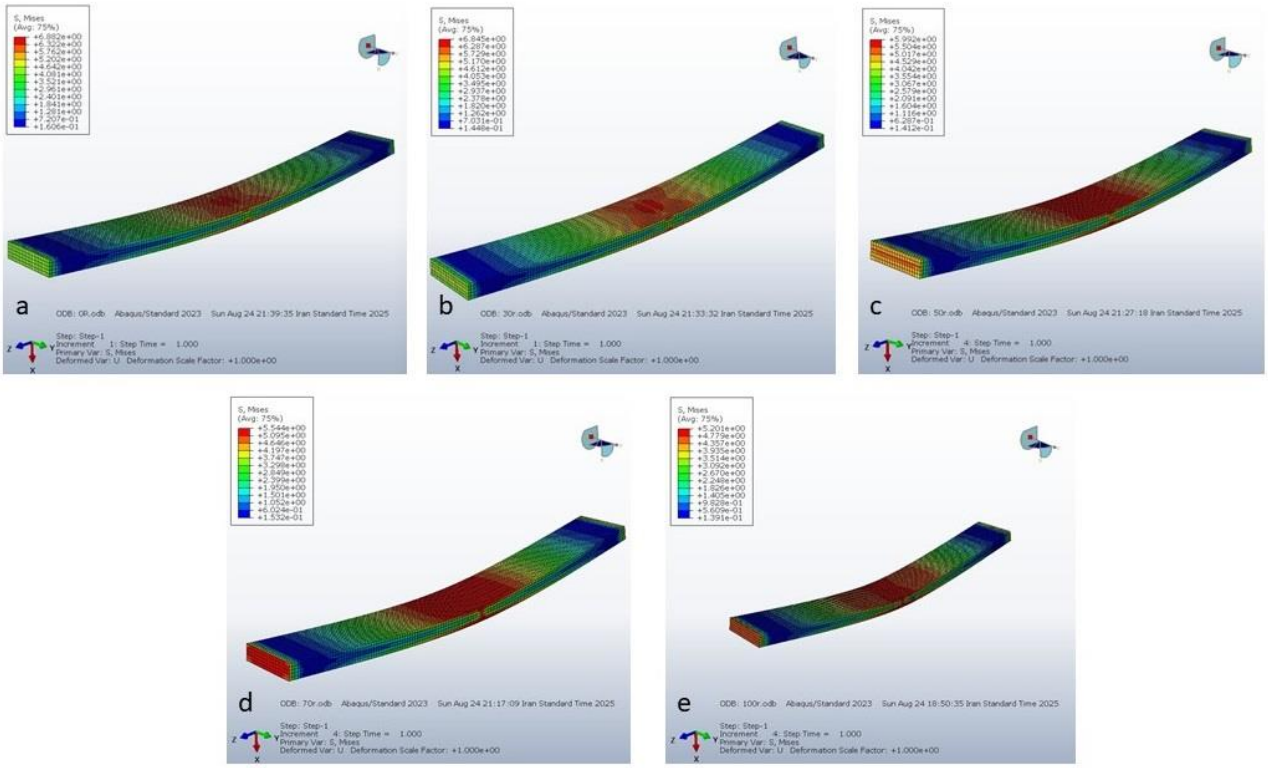


Figure 4: Von Mises stress results under a 10 N applied force for different material compositions: (a) 100% virgin, (b) 30% recycled, (c) 50% recycled, (d) 70% recycled, and (e) 100% recycled

Displacement

On the other hand, the displacement results (Figures 5 and 6) showed that with a force of 10 N, the displacement was about 3 to 4 mm, and with an increase in force to 20 N, this value reached about 5 to 10 mm. It is worth noting that for sensitive applications, the lower the displacement, the more suitable the structure will be; however, the stress behavior must also be considered at the same time.

In conclusion, it can be mentioned that although increasing the percentage of recycled polyethylene leads to a decrease in the Von Mises stress and an increase in the safety factor, the significant increase in displacement, especially at higher loads, creates limitations. Of course, it is suggested that in the future, researchers conduct more studies on quantum polymers and the effects of quantum dots on recycling, because quantum dots also need to be used in the development of various polymers. The addition of quantum dots to polymers is an effective way to enhance their optical and electronic properties. Quantum dots are nano-scale semiconductor particles with size-dependent luminescence and high optical efficiency. When embedded in a polymer matrix, they can improve functions such as light emission, electrical conductivity, and sensing while retaining the flexibility and processability of polymers. These polymer–quantum dot composites are widely used in optoelectronic devices, sensors, and biomedical applications. However, achieving uniform dispersion and strong interfacial bonding remains a key challenge for optimizing performance.

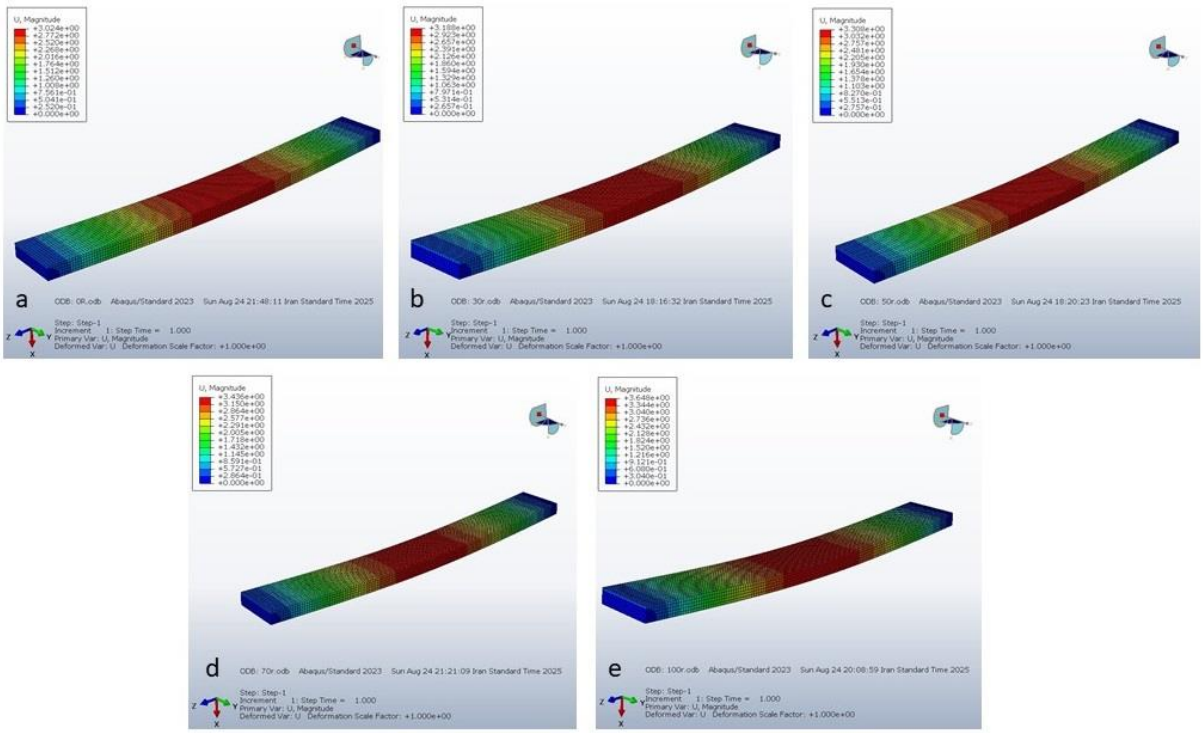


Figure 5: Displacement results under a 10 N applied force for different material compositions: (a) 100% virgin, (b) 30% recycled, (c) 50% recycled, (d) 70% recycled, and (e) 100% recycled

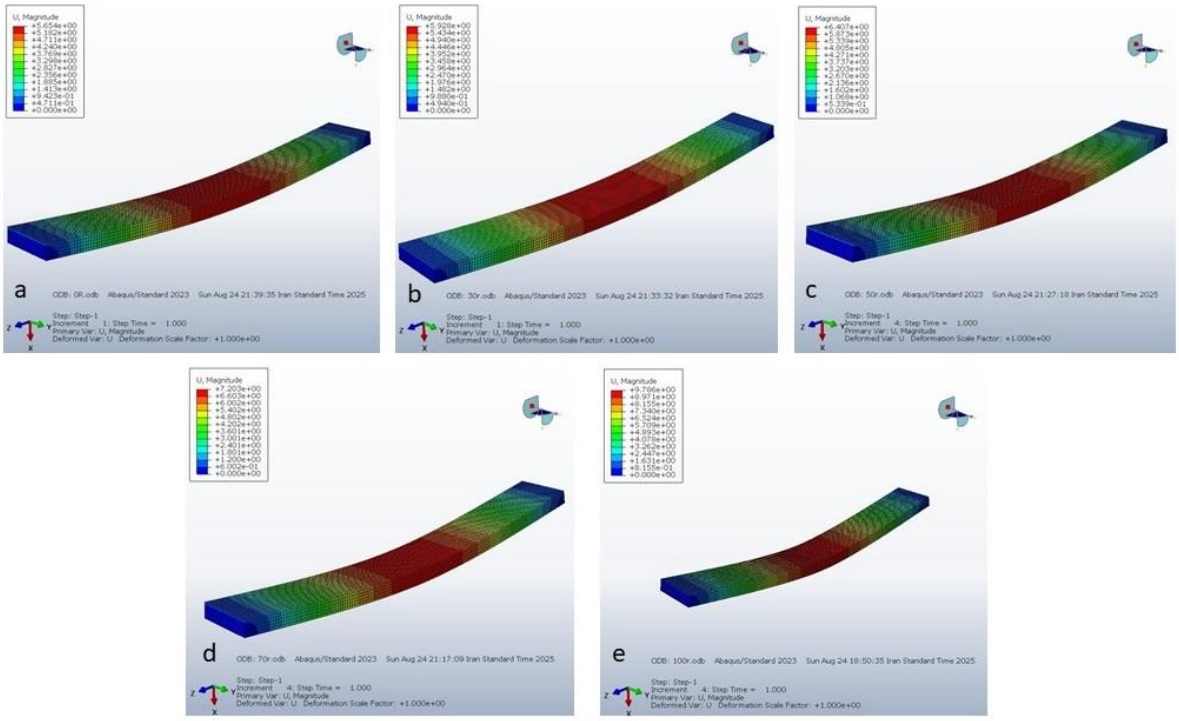


Figure 6: Displacement results under a 20 N applied force for different material compositions: (a) 100% virgin, (b) 30% recycled, (c) 50% recycled, (d) 70% recycled, and (e) 100% recycled

Plastic Equivalent Strain (PEEQ)

Figure 7 presents the PEEQ distributions for the compositions that exhibited non-zero plastic deformation at 20 N. Under a concentrated load of F = 10 N, all samples remained within the elastic region, with PEEQ = 0. When the load was increased to F = 20 N, the 0% and 30% recycled samples still exhibited purely elastic behavior (PEEQ = 0), whereas the 50%, 70%, and 100% recycled samples showed plastic deformation with PEEQ values of 0.0046, 0.0079, and 0.0190, respectively. These results indicate that increasing the recycled high-density polyethylene (rHDPE) content reduces the material stiffness and accelerates the onset of plastic deformation. This behavior can be attributed to the reduced molecular weight, lower chain entanglement density, and possible microstructural heterogeneity in recycled HDPE, which collectively decrease stiffness and promote earlier onset of plastic deformation under loading. Therefore, for applications where permanent deformation is undesirable, the proportion of rHDPE should be carefully limited [24].

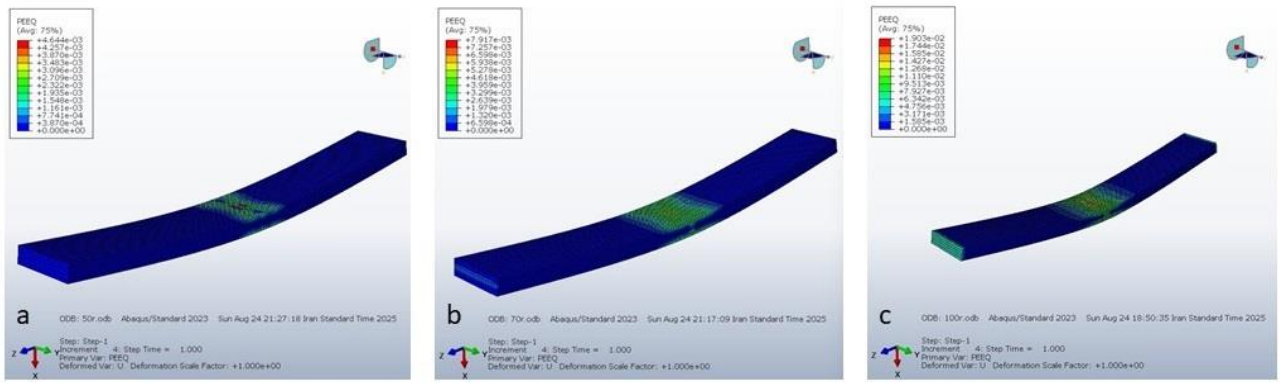


Figure 7: Plastic equivalent strain results under a 20 N applied force for different material compositions: (a) 50% recycled, (b) 70% recycled, and (c) 100% recycled

Conclusion

In this study, the mechanical behavior of polyethylene beams containing different percentages of recycled polyethylene was investigated using finite element analysis in Abaqus. The results demonstrated that increasing the recycled polyethylene content leads to a reduction in Von Mises stress capacity and overall stiffness, while simultaneously increasing plastic displacement and plastic strain. Under a lower applied load of 10 N, all specimens exhibited purely elastic behavior with negligible differences in plastic response. However, when the load was increased to 20 N, specimens with higher recycled content progressively entered the plastic region, exhibiting greater permanent deformation, whereas low-recycled-content specimens largely remained within the elastic range. The Plastic Equivalent Strain (PEEQ) results further confirmed this trend, showing that plastic deformation becomes more pronounced as recycled content increases, indicating a reduction in resistance to permanent deformation. This behavior is attributed to the altered mechanical integrity of the material with increasing recycled polyethylene content, which affects stiffness and deformation resistance. Among all investigated configurations, the specimen containing 30% recycled polyethylene demonstrated the most balanced performance, combining acceptable mechanical strength, limited deformation, and reduced use of virgin material. From an application perspective, this composition provides an optimal compromise between mechanical reliability and environmental sustainability. Therefore, it can be concluded that a 30% recycled polyethylene content is most suitable for semi-structural and moderately loaded applications, such as rigid packaging, general urban components, and similar engineering uses where both performance and material sustainability are required. Although a detailed life cycle assessment (LCA) was not performed in this study, the environmental benefit of incorporating recycled polyethylene can be qualitatively justified by the reduction in demand for virgin polymer production and the associated decrease in plastic waste accumulation. In general, mechanical recycling of polyethylene is considered an effective approach for lowering the environmental footprint of polymer-based materials by extending material life cycles and reducing landfill disposal. Future work will focus on experimental validation of the proposed composites and extended analysis under long-term aging, thermal cycling, fatigue, and impact loading conditions. These studies will further evaluate the durability and industrial applicability of recycled polyethylene-based composites.

Declarations Statements

The authors declared that they had no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Conflicts of Interest

The authors declared that there was no conflict of interest.

Data Availability Statement

All data can be requested and will be sent by corresponding author.

Authors' Contributions

Yasaman Ghasemi: Software simulation, 3D design model, first draft, and final approval.
Omid Ashkani: Supervision, final writing, and final approval.

AI-Assisted Technologies Statement

The authors declared that they used only used ChatGPT's artificial intelligence to improve grammar and correct the spelling of some words.

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